**QPCode** **Reg.No : .....**

**Name : .....**

# MAHATMA GANDHI UNIVERSITY, KOTTAYAM
## MGU-UGP (HONOURS) EXAMINATION
## FIFTH SEMESTER
## MG5DSEBCA307 - CROSS-PLATFORM APPLICATION DEVELOPMENT WITH DART AND FLUTTER
## (2024 ADMISSION ONWARDS)

**Duration: 2 Hours** **Maximum Marks: 70**

***Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)***

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

## Part A

Answer all questions. Each question carries 2 marks

1. What is the purpose of the abstract keyword in Dart? [U] / [CO1]
2. List any four key features of the Dart programming language [K] / [CO1]
3. Differentiate between ?? and ??= operators in Dart. [U] / [CO2]
4. State the difference between MaterialApp and CupertinoApp widgets. [U] / [CO3]
5. What is a Tween in Flutter animations? [K] / [CO4]

**[ 5 x 2 = 10 ]**

## Part B

Short Answer Questions

Answer any 5 questions. Each question carries 6 marks

6. Explain the concept of named constructors and mixins in Dart with examples. [U] / [CO1]
7. Compare and contrast the use of switch statements and if-else blocks for conditional flow in Dart, and demonstrate how to use iterable-specific loops to process a collection with a code snippet [U] / [CO1]
8. Describe the structure of a Dart function, and differentiate between positional optional parameters and named optional parameters, providing examples of how default values are assigned to each [U] / [CO1]
9. Explain the concept of Isolates in Dart. How do they differ from Futures in handling concurrent operations? [U] / [CO2]
10. Explain the basics of navigation and routing in a Flutter application, detailing the various techniques that can be utilized to pass data between different pages and widgets. [U] / [CO3]
11. How can you create responsive layouts in Flutter? Explain the use of LayoutBuilder and MediaQuery. [A] / [CO3]
12. Write the steps to perform manual JSON parsing in a Flutter application with a suitable example. [A] / [CO4]

**[ 5 x 6 = 30 ]**

## Part C

Essay Questions

Answer any 2 questions. Each question carries 15 marks

13. Compare Stateless and Stateful widgets, and explain the distinct roles of the Element and RenderObject in Flutter's architecture. Demonstrate the use cases for fundamental layout widgets like Row, Column, and Stack [A] / [CO2]

14. Explain the Provider pattern for state management in Flutter. How do you pass the state down the widget tree and update the UI efficiently? [A] / [CO3]

15. Describe how to fetch data from the internet using HTTP packages in Flutter. Explain the procedure to persist this fetched data locally using SQLite. [A] / [CO4]

**[ 2 x 15 = 30 ]**